CO2 - AT1

Analytical Problem Solving

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1) Count odd and Even Numbers in a singly Linked List.

Solution :

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Problem Statement :

Print all the odd and even counts present in the single Linked list having 10 elements:

77 → 88 → 22 → 11 → 66 → 27 → 55 → 65 → 18 → 10

Algorithm :

1) Start from the head node of the linked list.
2) Initialize two counters: odd count = 0 and evencount = 0
3) Traverse the list node by node until the current node becomes NULL.
4) At each node, check the value of data % 2 :
   - if data % 2 == 0, increment evencount by 1
   - Else, increment odd count by 1.
5) Move the pointer to the next node.
6) Repeat steps 4-5 until all nodes are visited.
7) After traversal, display the final odd count and evencount

Dry Run / Trace Table :-

Node visited	Data	Check (data % 2)	odd / Even	Odd count	Even count
1)	77	77 % 2 = 1	odd	1	0
2)	88	88 % 2 = 0	Even	1	1
3)	22	22 % 2 = 0	Even	1	2
4)	11	11 % 2 = 1	odd	2	2
5)	66	66 % 2 = 0	Even	2	3
6)	27	27 % 2 = 1	odd	3	3
7)	55	55 % 2 = 1	odd	4	3
8)	65	65 % 2 = 1	odd	5	3
9)	18	18 % 2 = 0	Even	5	4
10)	10	10 % 2 = 0	Even	5	5

Final result :-

- odd count = 5
- Even count = 5

2) Linear Search for Key K = 20 in a singly Linked List.

Solutions:

Problem statement:

Search linearly for a key element K = 20 in the key K = 20 in the single linked list. If found, Print "Yes"; else Print "NO"

List: 77 → 88 → 22 → 11 → 66 → 27 → 55 → 65 → 18 → 20

Algorithm:

1) Start from the head node of the linked list.

2) Set a flag variable found = 0 (false)

3) Traverse the list node by node until the current node becomes NULL.

4) At each node, compare the node's data with the key value K:

- If data == K, set found = 1 and stop the traversal immediately (element found).
- Else, move to the next node and continue.

5) After the traversal ends Check the flag:

- If found == 1, Print "Yes"
- If found == 0, Print "NO".

Dry Run / Trace Table

Step	Node Data	Compare with K=20	Result	Action.
1)	77	77=20?	NO	move to next node
2)	88	88=20?	NO	move to next node
3)	22	22=20?	NO	move to next node
4)	11	11=20?	NO	move to next node
5)	66	66=20?	NO	move to next node
6)	27	27=20?	NO	move to next node
7)	55	55=20?	NO	move to next node
8)	65	65=20?	NO	move to next node
9)	18	18=20?	NO	move to next node
10)	10	10=20?	NO	Node = NULL, traversal ends

Since no node matched the key, found remains 0.

Final Result

output: NO (key 20 is not present in the list)